

Practice exam

Python Programming for Business Intelligence

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Exam setup

Be in a quiet area where you cannot be disturbed

Do not have anything extra on the tables besides your computer and an ID

Have only Teams and the provided exam Jupyter Notebook open

Any suspicion of cheating (having extra notes, AI tools, etc.) can result in me ending the exam and requiring a retake

Common issues

Arrive on time! Arriving late results in time deducted from the exam itself, but the number of questions stays the same

Make sure you can share your screen on Teams (check your own computer's permissions, test with a friend...)

Make sure your Teams works on the computer you plan on using. Do not join on a phone.

During the exam itself

Ask for clarification if something is unclear

In the exam, I will ask the questions verbally. There will be no slides with the question. In some questions, I might share my screen and show you code or a visualization

If I am hinting you something is not correct or try to make you answer faster, it is for your own benefit. I'm giving you a chance to improve your answer and not run out of time

At the end I will give you the final score. There is no negotiating the result, so please don't start complaining or requesting more points. I grade the exam without considering your course score. Even if you end up with 79/80, your final grade is 3.

Doing the practice exam outside the session

If you are going through this and did not participate in the exercise session, here are some additional instructions:

- Practice the exam first without using any answers or materials visible
- Use a timer and aim to answer each question within the suggested time given in the slide

This exam gives you an idea what to expect

... but do not expect the exam questions to be exactly like the ones provided here

The exam questions will vary throughout the day and week. Thus, there is no benefit of being the last to go as the questions the previous people were asked might not appear in your exam

Practice exam

Question 1

Question: Tell me about the different data structures in Python we used during the course, both the ones within Python and from the two main data science libraries

Suggested time: max 2 min

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Question: Tell me about the different data structures in Python we used during the course, both the ones within Python and from the two main data science libraries

A good answer would cover dictionaries, lists, tuples, numpy arrays and pandas DataFrames

Suggested time: max 2 min

Question 2

Question: Walk me through what you should do when you get your hands on a new dataset?

Suggested time: max 2 min

Question 2

Question: Walk me through what you should do when you get your hands on a new dataset?

A good answer would describe all aspects of EDA

Suggested time: max 2 min

Question 3

Question: Let's open Jupyter. Show me how you would define a list, dictionary and tuple. You can choose what to put in them. Show how you access elements within them

Suggested time: max 3 min

Question 4

Question: We covered a variety of different data sources, such as file formats, and systems that store data during the course. Can you go through the ones you remember and briefly tell about them?

Suggested time: max 3 min

Question 4

Question: We covered a variety of different data sources, such as file formats, and systems that store data during the course. Can you go through the ones you remember and briefly tell about them?

A good answer would mention databases, flat files (csv, JSON), APIs and explain what they are and what benefits/weaknesses they have

Suggested time: max 3 min

Question 5

Question: Let's look at this piece of code. Explain to me what you see and what is happening here

Suggested time: max 2 min

```
url = "https://api.open-meteo.com/v1/forecast"
params = {
    "latitude": 60.17,
    "longitude": 24.94,
    "daily": "temperature_2m_max,temperature_2m_min",
    "timezone": "Europe/Helsinki"
}

response = requests.get(url, params=params)
weather = response.json()

print("Status code:", response.status_code)
```

Question 6

Question: What is the different between supervised and unsupervised machine learning? Can you describe practical business applications of both?

Suggested time: max 2-3 min

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Question: What is the different between supervised and unsupervised machine learning? Can you describe practical business applications of both?

A good answer would describe the main differences and then provide at least 2 examples of applications of both supervised and unsupervised ML

Suggested time: max 2-3 min

Feedback

After the exam, I will briefly note how it went and stating how many points you got